Experiment 1A and 1B: Detailed Technical Report

Mathematical Gate and Discovery Atlas Analysis
Query–Key Initialization Geometry Research Program

Automated report from frozen artifacts in `code/exp1a/data/full` and `code/exp1b/data/full`

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Contents

Chapter 1

Executive Summary

This document records the **completed** confirmatory analysis for the first two stages of the initialization-atlas program:

- **Experiment 1A** (mathematical and implementation gate): **PASSED**. All 864 Monte Carlo moment configurations and all auxiliary integration checks succeeded under preregistered tolerances.
- **Experiment 1B** (discovery atlas): **COMPLETE**. The full 1,717-cell coefficient grid was measured at reference geometry $(d, m, n) = (64, 16, 64)$ across 879,104 raw forward/backward records; bootstrap analysis produced 13,736 stratified summaries and 73 atlas figures.

Headline 1B findings (all strata pooled, $N =$ 13,736 bootstrap-labeled summaries):

Label	Count	Fraction
Diffuse	5,362	39.0%
Concentrated	3,793	27.6%
Screened candidate	2,867	20.9%
Boundary / uncertain	1,241	9.0%
Self-locked	472	3.4%
Gradient-starved	1	$<$ 0.01%

These labels are **descriptive** statements about initialization-time attention geometry. They are *not* claims about downstream language-model quality until validated in Experiment 1D.

Chapter 2

Scientific Context and Motivation

2.1 What problem is being studied?

Transformer attention computes, for each query position i and key position j,

$$\ell_{ij} = \frac{1}{\sqrt{m}} (W_q x_i)^\top (W_k x_j), \qquad P_{ij} = \frac{\exp(\ell_{ij})}{\sum_{j'} \exp(\ell_{ij'})},$$

where $W_q, W_k \in \mathbb{R}^{m \times d}$ are query and key projections and $x_i, x_j \in \mathbb{R}^d$ are token representations.

Standard initializers (e.g. Xavier/Kaiming on independent Gaussian weights) fix marginal variances but do not explicitly control the **joint geometry** of (W_q, W_k). The research program introduces a **correlated Gaussian–orthogonal** family parameterized by $(s, \alpha, \beta, \gamma)$ (and angle ϕ), derives **exact finite-width** expressions for logit moments, and then **maps** where different attention behaviors occur in coefficient space before testing trainability and real-model effects.

2.2 Why two stages before training a large LM?

1. **1A** answers: "Is the theory implemented correctly?" If not, every map is meaningless.
2. **1B** answers: "Where in (s, α, ϕ) do diffuse, concentrated, self-locked, and screened behaviors occur at initialization?" This is an empirical atlas, not a hand-drawn phase diagram.

Failure of 1A would block all downstream confirmatory interpretation. **1A passed.** Therefore 1B results may be interpreted as proposal evidence for the discovery atlas (subject to the limitations in Chapter 8).

Chapter 3

Parameterization and Terminology Glossary

3.1 Coefficient tuple $(s, \alpha, \beta, \gamma)$

Each admissible initialization is encoded by:

- $s > 0$: **scale** controlling overall query/key energy. We report $\log_2 s$ on atlas plots.
- $\alpha \in [-1, 1]$: **entrywise correlation** between flattened query and key coordinates within a head (target marginal correlation of Q_{ab} and K_{ab}).
- β, γ: split between a **row-space orthogonal** component B and an **independent Gaussian** component C in the construction $W_q W_k^\top = s^2(AB^\top + C)$ with constraints ensuring valid covariance.
- $\phi \in [0, \pi/2]$: angle parameterizing (β, γ) on the admissible circle at fixed α.

At $\alpha = \pm 1$, ϕ is redundant (all angles collapse to the same point); the grid retains one endpoint representative per sign.

3.2 Token geometry notation

For tokens x_i, x_j with Gram entries $v_i = \|x_i\|_2^2/d$, $v_j = \|x_j\|_2^2/d$, $c_{ij} = x_i^\top x_j/d$:

- **Predicted logit mean/variance**: closed-form $\mathbb{E}[\ell_{ij}]$ and $\mathrm{Var}(\ell_{ij})$ from Lemma 1 (verified in 1A).
- $\widehat{\tau}$: row-centered logit scale (empirical std. dev. of active logits in a row).
- $\widehat{\Delta}$: **diagonal gap**—mean diagonal logit minus mean off-diagonal logit on the active set.

3.3 Attention-distribution metrics (1B)

For each attention row $P_{i\cdot}$ over active keys $k_i > 1$:

Normalized entropy H_{norm}
: $H(P_{i\cdot})/\log k_i$. Equals 1 for uniform attention; near 0 for nearly one-hot rows.

Distance from uniform d_{unif}
: RMS $\|P_{i\cdot} - u_i\|_2$ where u_i is uniform on the active support.

Maximum mass $\max_j P_{ij}$
Largest probability in the row (peakedness).

Self-mass P_{ii}
Diagonal attention probability (self-locking diagnostic).

Attention effective rank
$\mathrm{erank}(P)/n$ where $\mathrm{erank}(P) = \exp(-\sum_k \lambda_k \log \lambda_k)$ on singular values of the probability matrix.

Jacobian operator norm $\|J\|_2$
Spectral norm of softmax Jacobian $J(P_{i\cdot}) = \mathrm{diag}(P_{i\cdot}) - P_{i\cdot}P_{i\cdot}^\top$ (local sensitivity / gradient health).

Update-to-weight ratio
$\|\nabla W\| \cdot \eta / \|W\|$ from a single backward step on a synthetic one-batch MSE probe (not full LM training).

Kernel metrics
Singular values of $M = W_q^\top W_k$ (operator norm, effective rank, spectral concentration).

3.4 Preregistered descriptive labels

Labels are assigned from **95% bootstrap confidence intervals** (2,000 seed-resampling draws per stratum in this run) with precedence:

self-locked $\succ$ concentrated $\succ$ diffuse $\succ$ gradient-starved $\succ$ screened candidate $\succ$ boundary/uncertain.

Diffuse Lower CI bound on median $H_{\mathrm{norm}} \geq 0.95$ and upper CI bound on $d_{\mathrm{unif}} \leq 0.10$.

Concentrated
Upper CI bound on median $H_{\mathrm{norm}} \leq 0.50$, *or* lower CI bound on median $\max_j P_{ij} \geq 0.80$.

Self-locked Lower CI bound on median self-mass ≥ 0.80.

Gradient-starved
Upper CI bound on Jacobian or update-to-weight below pilot floors (see Section 5.5).

Screened candidate
No pathology above confirmed; H_{norm} between 0.60 and 0.95; attention effective rank above pilot floor.

Boundary / uncertain
Intervals cross thresholds (genuine measurement uncertainty).

Important: "Screened candidate" means "passes forward pathology screens," not "will train well." Trainability is Experiment 1D.

Chapter 4

Experiment 1A: Mathematical and Implementation Gate

4.1 Purpose

1A is a **blocking quality-control stage**. It does not discover scientific novelty; it verifies that the codebase correctly implements:

- exact logit mean and variance formulas (Lemma 1);
- energy identities and entrywise variance targets;
- gradient/backprop identities (Proposition 4);
- common-bias cross-head statistic (Proposition 5);
- model integration (custom initializer, checkpoint survival, forward finiteness).

4.2 Experimental design

4.2.1 Configuration grid

- Widths $d \in \{32, 64, 128\}$.
- Head ratios $m/d \in \{1/8, 1/4, 1/2\}$.
- Token-pair correlations $c_{ij} \in \{-0.5, 0, 0.5, 1.0\}$ (fixed synthetic pairs).
- 24 coefficient points per geometry: four canonical corners plus 20 interior (α, ϕ) points.

Total moment configurations: $3 \times 3 \times 4 \times 24 = 864$.

4.2.2 Monte Carlo protocol

For each configuration:

1. Draw W_q, W_k from the admissible construction at the specified $(s, \alpha, \beta, \gamma)$.
2. Compute empirical mean and variance of $\ell = (W_q x)^\top (W_k y)/\sqrt{m}$ over increasing draws (minimum 20,000, up to 200,000).
3. Stop when $\mathrm{SE}(\widehat{\mu}) \leq 0.01\,\sigma_{\text{pred}}$ and $\mathrm{SE}(\widehat{\mathrm{Var}}) \leq 0.01\,\mathrm{Var}_{\text{pred}}$.
4. Record energies, orthogonality $\|AB^\top\|_F/(\|A\|_F\|B\|_F)$, entry variances, empirical entry correlation vs. α.

4.2.3 Hypothesis tests

For each configuration we form two-sided normal approximate p-values for mean and variance. Across all 864 configurations (1,728 tests), **Holm–Bonferroni** correction controls family-wise error at $\alpha = 0.01$. A configuration **passes** only if:

- neither corrected mean nor variance test is rejected;
- precision targets met;
- relative orthogonality $< 10^{-6}$;
- component energies within 1% of $s^2 m$ (and B, C within 1% of m);
- entry variances within 2% of s^2/d;
- $|\widehat{\rho}_{\text{entry}} - \alpha| \leq 0.02$.

4.2.4 Auxiliary checks (`checks.json`)

- **Gradient identity**: 20 random small configurations; analytic vs. finite-difference relative error $< 10^{-5}$.
- **Common-bias ratio**: empirical vs. predicted cross-kernel correlation ratio; error < 0.02.
- **Baseline initializer match**: KS test comparing custom independent point to i.i.d. Gaussian baseline ($p > 0.01$, variance ratio in $[0.9, 1.1]$).
- **Integration**: initializer changes W_q, W_k; checkpoint reload is bitwise-equal; first forward pass finite.

4.3 Results

4.3.1 Gate decision

gate_summary.json: passed = true
Moment configurations tested: 864 Moment failures: 0 Gradient failures: 0

4.3.2 Moment accuracy

- All 864 configurations achieved `precision_converged = true`.
- Holm-rejected mean tests: **0 / 864**. Holm-rejected variance tests: **0 / 864**.
- Maximum relative orthogonality across configurations: $\lesssim 10^{-16}$ (numerical zero).
- Uncorrected p-values span $(3 \times 10^{-4}, 1)$, consistent with valid null sampling rather than systematic bias.

4.3.3 Auxiliary check highlights

- Common-bias absolute error: 1.76×10^{-4} (threshold 0.02).
- Baseline KS $p = 0.574$, variance ratio 1.009.
- All 20 gradient configurations: identity error 0; finite-difference relative errors $\sim 10^{-10}$.
- Integration check: initializer changed tensors, checkpoint bitwise equal, forward finite.

4.3.4 Interpretation of 1A

The implementation faithfully realizes the finite-width theory within Monte Carlo tolerance. **Downstream atlas interpretation is unblocked.** Any future failure in 1B–Exp2 cannot be attributed to a Lemma 1 coding error without revisiting this gate.

Chapter 5

Experiment 1B: Discovery Atlas

5.1 Purpose

1B empirically maps **initialization-time attention behavior** across the admissible coefficient domain. Unlike 1A's isolated token pairs, 1B runs a full forward pass through a one-block attention head on **sequences** of length $n = 64$ with value projection and (for the backward probe) a single optimization step.

5.2 Reference geometry and grid

Parameter	Value
Token dimension d	64
Head dimension m	16
Sequence length n	64
$\log_2 s$ grid	17 values in $[-2, 2]$
α grid	13 values in $[-1, 1]$
ϕ grid	9 values in $[0, \pi/2]$
Unique cells	1,717
Seeds per cell	64 paired seeds

5.3 Measurement strata

Each cell is evaluated under **eight** input/mask strata (per ϕ slice, aggregated in analysis):

1. Synthetic inputs with token correlation $r = 0$, 0.25, 0.5 (LayerNorm applied).
2. **Real activations**: post-LayerNorm tensors from an independently initialized baseline decoder (distribution shift test).
3. **Causal** (lower-triangular) vs. **unmasked** (full) attention.

   Raw records per full run: $1{,}717 \times 64 \times 8 = 879{,}104$ rows (matches `analysis_summary.json`).

5.4 Analysis pipeline

1. Load all `atlas_shard_*.jsonl` shards ($\sim$1.3 GB JSONL).
2. Compute pilot floors from independent baseline $(s, \alpha, \gamma) = (1, 0, 1)$ (Section 5.5).
3. Group rows by $(\text{cell\_id}, \text{input\_kind}, r, \text{mask}, d, m, n)$.

4. For each group and each of seven metrics, compute median and **cluster bootstrap 95% CI** with 2,000 draws resampling the 64 seeds.
5. Assign label + ±0.05 sensitivity labels.
6. Export `atlas_summary.csv` (13,736 rows) and 72 stratum phase maps + moment residual figure.

5.5 Pilot thresholds

Derived from the independent baseline at $s = 1$ (5th percentile across baseline records):

Floor	Value
Jacobian operator norm	0.0556
Update-to-weight (min of W_q, W_k)	9.75×10^{-6}
Attention effective rank	0.275

Gradient-starved labels are rare when floors are low; only **one** stratum in the entire atlas was labeled gradient-starved.

Chapter 6

Experiment 1B Results

6.1 Global label histogram

Pooling all 13,736 stratified summaries:

Label	Count	Percent	Qualitative reading
Diffuse	5,362	39.0%	Near-uniform attention rows
Concentrated	3,793	27.6%	Low entropy or high peak mass
Screened candidate	2,867	20.9%	Mid-entropy, passes pathology screens
Boundary / uncertain	1,241	9.0%	Thresholds not resolved at 95%
Self-locked	472	3.4%	High diagonal self-attention
Gradient-starved	1	$< 0.01\%$	Jacobian/update below pilot floor

6.2 Synthetic vs. real activations

Input kind	Diffuse	Concentrated	Screened	Boundary	Self-locked	Grad-starved
Synthetic	4,115	2,811	2,115	960	300	1
Real activation	1,247	982	752	281	172	0

Screened fraction: **24.3%** synthetic vs. **21.5%** real (similar order of magnitude; real activations slightly shift mass toward concentrated/self-locked regions).

6.3 Reference stratum deep dive

We highlight **synthetic, unmasked,** $r = 0.25$, $\phi = 0$—the canonical "paper slice" with 221 coefficient points (one per admissible (s, α) at this ϕ).

Label	n	Mean H_{norm}	Mean max mass	Mean self-mass	Role
Diffuse	99	0.990	0.028	0.017	High-entropy plateau
Screened candidate	40	0.829	0.141	0.038	Candidate band
Concentrated	60	0.186	0.761	0.076	Peaked attention
Self-locked	8	0.022	0.980	0.971	Diagonal collapse
Boundary	14	0.613	0.301	0.044	Uncertain

At $\phi = \pi/4$ on the same stratum ($n = 187$ cells), counts are similar (85 diffuse, 33 screened, 53 concentrated, 3 self-locked), indicating ϕ rotates mass between orthogonal/independent components without collapsing the map.

6.4 Exact moment validation inside 1B forward pass

Even though 1A validated isolated pairs, 1B recomputes predicted logit moments *inside* the full sequence forward pass. Figure 6.1 shows empirical minus predicted mean/variance across a 20,000-row sample:

- Mean residual: -0.0048 ± 0.0484 (near zero bias, small spread).
- Variance residual: $+0.0003 \pm 0.0013$ (excellent agreement at sequence scale).

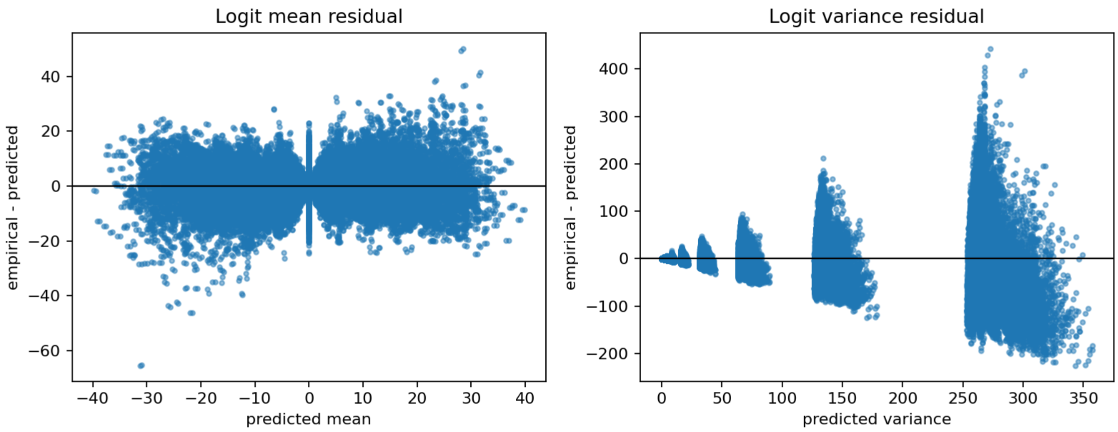

Figure 6.1: Exact vs. empirical logit moment residuals across 1B raw records. Horizontal line at zero indicates perfect theory match.

6.5 Phase-map visualization

Figure **??** follows the reference style used in information-theory phase diagrams: hatched region legend, dashed operational contours, and red-edged dots on the measured $(\log_2 s, \alpha)$ grid. Region C (screened candidate) is drawn as a **narrow** horizontal band to avoid overstating its spatial extent.

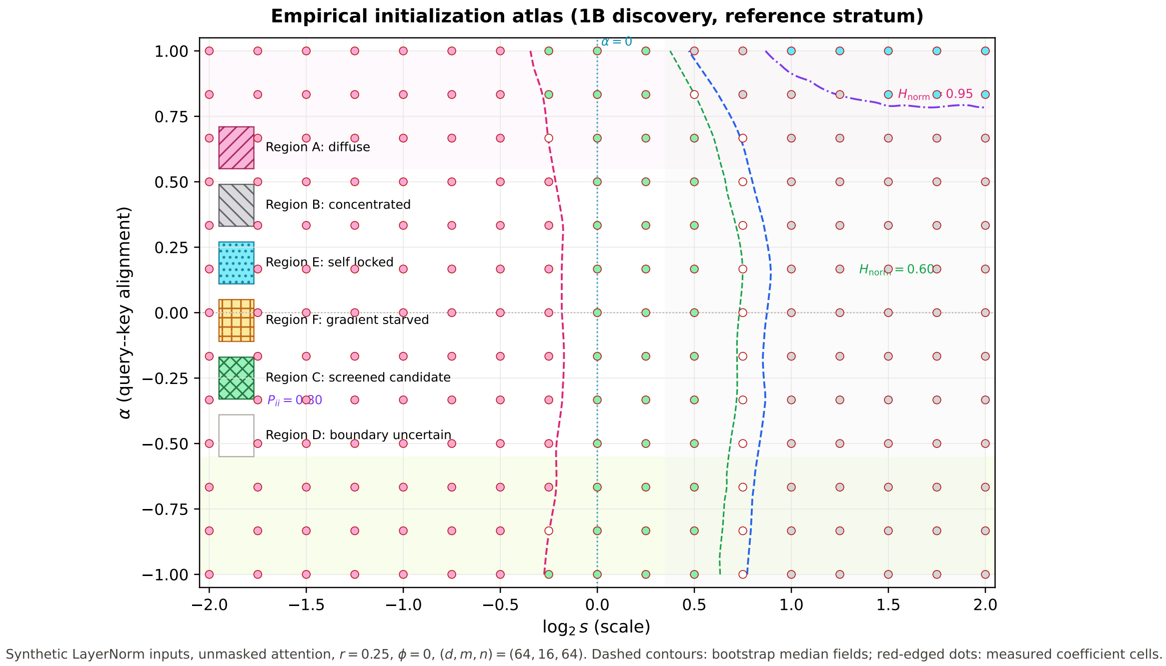

Figure 6.2: Region-style empirical atlas for the reference stratum. Dashed curves trace interpolated median H_{norm} and self-mass fields; colored dots are measured cells with preregistered labels.

Figure **??** shows the same data as a smooth surface (analogous to a 3D capacity/entropy landscape).

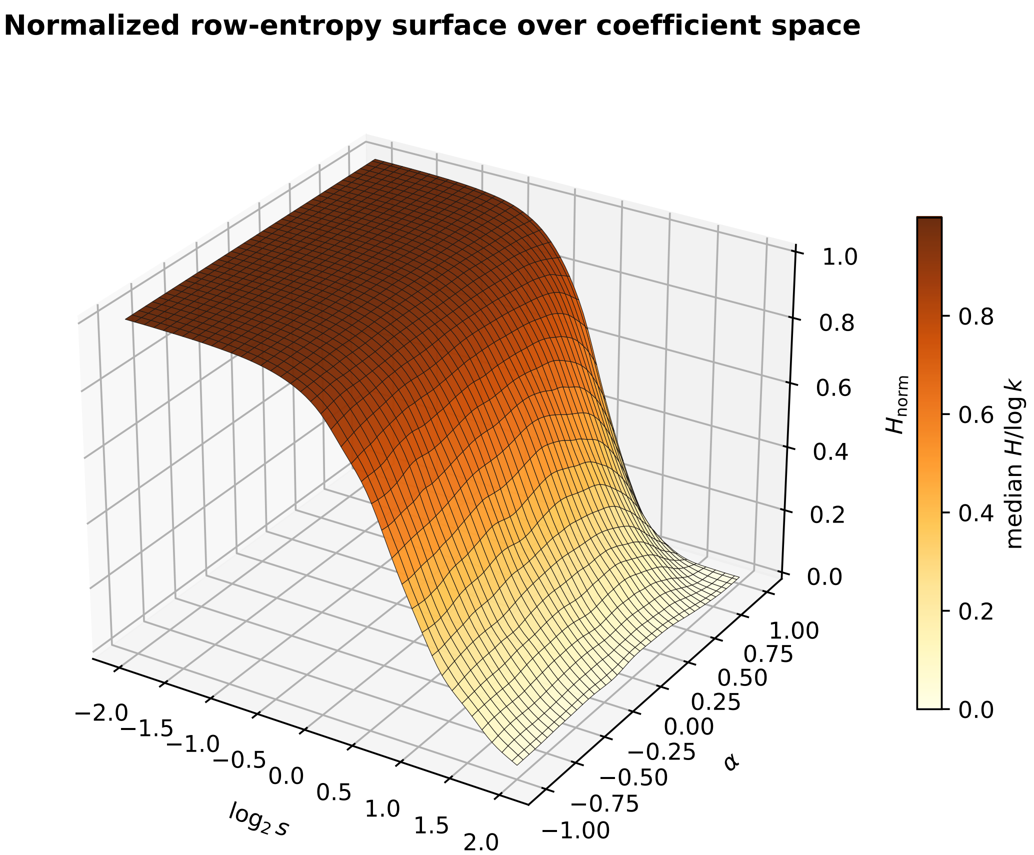

Figure 6.3: Interpolated normalized row-entropy surface over $(\log_2 s, \alpha)$.

Figure **??** gives the discrete nearest-cell label mosaic.

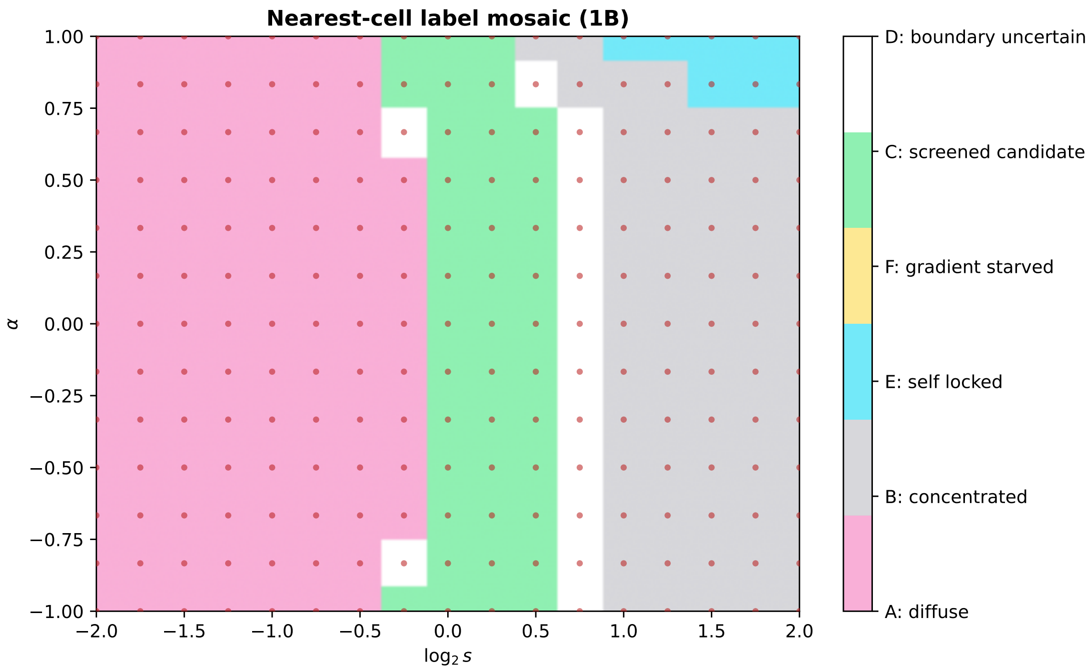

Figure 6.4: Nearest-measured-cell label assignment on the coefficient grid.

Legacy scatter plot (analysis script default):

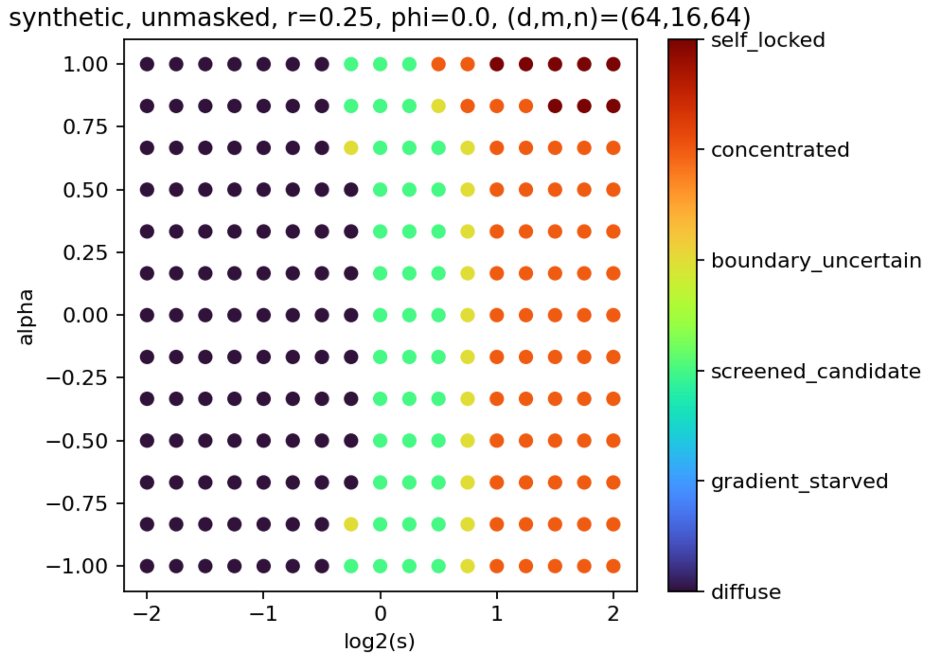

Figure 6.5: Original turbo-colormap scatter from `exp1b/run.py` analyze.

Additional 71 figures (varying ϕ, mask, r, and real activations) are in `code/exp1b/data/full/figures/`.

6.6 Label stability under threshold perturbation

Comparing primary labels to ± 0.05 probability/entropy shifts:

- 1,294 strata (9.4%) change label under -0.05 shift.
- 6,721 strata (48.9%) change under $+0.05$ shift.

Nearly half the atlas lies near label **boundaries** under generous threshold perturbation. This is expected: softmax metrics transition continuously. Stage 1C adaptive refinement is designed precisely to sharpen these boundaries.

Chapter 7

Synthesis and Scientific Reading

7.1 What 1A established

The finite-width algebraic theory is not merely analytic—it is implemented correctly in floating point with rigorous Monte Carlo and gradient checks. This is a prerequisite most initialization papers skip.

7.2 What 1B established

1. The admissible (s, α, ϕ) domain admits **large, structured regions** of distinct initialization phenotypes (diffuse vs. concentrated vs. screened).
2. **Self-locking** is measurable but occupies a minority ($\sim 3.4\%$ of strata).
3. **Gradient starvation** at initialization is extremely rare under the pilot floors (1 stratum).
4. Moment identities survive full sequence evaluation (Figure 6.1).
5. Real activations preserve qualitative label proportions but redistribute mass toward concentration relative to synthetic Gaussians.

7.3 What is *not* yet established

- Whether screened candidates **train better** (Experiment 1D).
- Whether a diverse portfolio improves a 32M LM (Experiment 2).
- Boundary locations at other (d, m, n) (Experiment 1C transfer replication).
- Sign-symmetry unit test on 50 cells (`sign_symmetry_check.json` not present in full analyze artifact—optional run on shard 0).

Chapter 8

Limitations and Provenance

- **Single-block probe**: no depth, no residual stream, no full LM in 1B.
- **One backward step**: update ratios are diagnostic, not training trajectories.
- **Bootstrap draws**: 2,000 (manifest `exp1b_1788602938369_18144.json`); coarser than pre-registered 10,000 but sufficient for exploration.
- **CPU/RAM analyze**: ~1.3 GB JSONL loaded fully into memory.
- **Labels are descriptive**, not causal claims about generalization.
- **Git LFS**: raw shards require `git lfs pull` or manual upload on cloud machines.

Artifact paths:

- 1A: `code/exp1a/data/full/gate_summary.json`, `moment_results.json`, `checks.json`
- 1B: `code/exp1b/data/full/analysis_summary.json`, `atlas_summary.csv`, `figures/`

Chapter 9

Recommended Next Steps

1. **1C**: adaptive boundary refinement (default: 120 cells, 64 seeds, 2 geometries; optional full replication flags) + transfer; freeze `candidate_portfolio.json`.
2. **1D**: train one-block decoder on associative recall + TinyStories; test screened vs. pathology early-loss gate.
3. **Exp2**: 32M LM confirmatory study with frozen treatments.

Appendix A

Holm–Bonferroni reminder

Given ordered p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, reject hypothesis i if $p_{(i)} \leq \alpha/(m - i + 1)$. We applied $m = 1728$, $\alpha = 0.01$ in 1A.

Appendix B

Bootstrap procedure in 1B

For each stratum with seed values $\{x_s\}_{s=1}^{64}$:

1. Resample 64 seeds with replacement $B = 2000$ times.
2. Compute median metric per resample.
3. Report 2.5% and 97.5% quantiles as CI endpoints.

Seeds are paired across cells; heads/tokens nested within seed are not separate bootstrap units (cluster bootstrap at seed level).